Nick (Seungheon) Oh

nick.sh.oh@gmail.com · nick-sh-oh.github.io · Google Scholar

EDUCATION

Apr 2022 — Sept 2023 | MSc Data Science and Artificial Intelligence, University of London | London
Distinction

Sept 2016 — Jun 2020 | BSc Politics and Economics, London School of Economics and Political Science (LSE) | London
Upper Second Class Honours [Cumulative GPA: 3.5-7/4.0]

SELECTED PUBLICATIONS

Oh, N., & Gobet, F. (2026). Small Foundation Models of Human Cognition and Behaviour. *The 3rd Conference on Language Modeling (**COLM'26**)* [Paper]

Gobet, F., Bennett, D., & **Oh, N.** (2026). Discrimination Nets in Cognitive Science. *Thirteenth Annual Conference on Advances in Cognitive Systems (**CogSys'26; Oral**)*

Oh, N., & Gobet, F. (2025). Monitor-Generate-Verify (MGV): Formalising Metacognitive Theory for Language Model Reasoning. *Workshop on Foundations of Reasoning in Language Models @ the 39th Annual Conference on Neural Information Processing Systems (**FoRLM Workshop, NeurIPS'25**)* [Paper]

CONFERENCE PROCEEDINGS

Oh, N., & Gobet, F. (2026). Small Foundation Models of Human Cognition and Behaviour. *The 3rd Conference on Language Modeling (**COLM'26**)* [Paper] [Poster]

Gobet, F., Bennett, D., & **Oh, N.** (2026). Discrimination Nets in Cognitive Science. *Thirteenth Annual Conference on Advances in Cognitive Systems (**CogSys'26; Oral**)*

Oh, N., Vrakas, G. D., Brooke, S. J., Morinière, S., & Duke, T. (2025). PETLP: A Privacy-by-Design Pipeline for Social Media Data in AI Research. *Proceedings of the 8th AAAI/ACM Conference on AI, Ethics, and Society (**AIES'25**)* [Paper] [Poster]

Oh, N. (2024). Sentibank: A Unified Resource of Sentiment Lexicons and Dictionaries. *Proceedings of the 18th International AAAI Conference on Web and Social Media (**ICWSM'24**)* [Paper] [Poster]

WORKSHOPS

Oh, N., & Jin, H. (2026). Reliability, Faithfulness, and the Limits of Post-hoc Explanations of Opaque Scientific Models. *Philosophy of Machine Learning Workshop @ the 43rd International Conference on Machine Learning (**PhilML Workshop, ICML'26**)* [Paper] [Poster]

Oh, N., & Gobet, F. (2025). Monitor-Generate-Verify (MGV): Formalising Metacognitive Theory for Language Model Reasoning. *Workshop on Foundations of Reasoning in Language Models @ the 39th Annual Conference on Neural Information Processing Systems (**FoRLM Workshop, NeurIPS'25**)* [Paper] [Poster]

Oh, N. (2025). Before you <think>, monitor: Implementing Flavell's metacognitive framework in LLMs. *Workshop on the Application of LLM Explainability to Reasoning and Planning @ Conference on Language Modeling (**XLLM-Reason-Plan Workshop, COLM'25**)* [Paper] [Poster]

Oh, N., & Gobet, F. (2024). System 1.5: Designing Metacognition in Artificial Intelligence. *System 2 Reasoning at Scale Workshop @ the 38th Annual Conference on Neural Information Processing Systems (**Sys2-Reasoning Workshop, NeurIPS'24; Spotlight**)* [Paper] [Poster] [Spotlight Presentation (Video)]

Oh, N. (2024). In Defence of Post-hoc Explainability. *Interpretable AI: Past, Present and Future Workshop @ the 38th Annual Conference on Neural Information Processing Systems (**IAI Workshop, NeurIPS'24**)* [Paper] [Poster]

WORKING PAPERS

Thinking Without Words: The Representational Geometry of Latent Reasoning Trajectories, with Cho, Y., Lee, M., Joo, S. & Lee, A.

TL;DR: Tests whether purely latent chains of thought can sustain long-horizon reasoning by characterising the representational geometry of their trajectories.

The Ideological Subspace of Large Language Models: Angular Steering of Political Representations, with Cho, Y. & Gobet, F.

TL;DR: Locates an ideological subspace in LLMs, characterises its structure, and uses angular steering to shift models' political positions.

Do Models of the Mind Converge? Representational Analysis of Neural Networks fitted on Human Behaviour, with Gobet, F.

TL;DR: Traces how representations in small foundation models of human cognition evolve with LoRA rank and data scale, and whether they converge across model sizes and families.

Mapping the Architecture–Task Space of Neural Network Models of Human Behaviour

TL;DR: A survey charting which neural network architectures have been trained on which behavioural tasks.

High-Intensity Ambivalence: Beyond Acceptance and Resistance to Generative AI in Occupational Communities (in review), with Humphrey, J. & Ross, D.

TL;DR: Argues that occupational communities' responses to generative AI are better understood as high-intensity ambivalence than as acceptance or resistance.

HONOURS, GRANTS & AWARDS

2026 — **$10,000 data grant**, Cookiy Labs

2026 — **Oral presentation**, CogSys'26, for *Discrimination Nets in Cognitive Science*

2026 — **IEEE TCPAMI Art Award** and Spotlight presentation, Art Gallery @ CVPR'26, for *artefact(s): LeNet-1*

2024 — **Spotlight presentation**, Sys2-Reasoning Workshop @ NeurIPS'24, for *System 1.5: Designing Metacognition in Artificial Intelligence*

PROFESSIONAL EXPERIENCE

Oct 2026 — Present — Researcher, London School of Economics and Political Science (LSE) — London

Visiting researcher at CPNSS (Centre for Philosophy of Natural and Social Science), advised by Professor Fernand Gobet.

Jan 2023 — Present — Researcher, socius labs — London

Founded socius labs, an independent research lab supported by the London School of Economics (LSE) and LSE Generate, bridging human sciences and artificial intelligence through two parallel research streams: applying insights from human sciences to enhance AI systems, and leveraging AI as a platform to run human theories at scale.

Mar 2025 — Sept 2026 — Defence AI researcher, Ministry of National Defence — Republic of Korea

Assigned to the S/C3 (Command, Control & Communications) Capability Requirements Branch, Training and Doctrine Command (TRADOC); on assignment from the Military Science and Technology Research Unit, Republic of Korea Army Headquarters [received TRADOC Commander's commendation for outstanding performance]

Nov 2021 — Oct 2022 — Junior NLP engineer, Numen Capital — London

Contributed to Knowsis research team extracting trading signals from non-traditional data sources (social media) through NLP and machine learning. Built transformer-based ESG (Environment, Social, Governance) tweet classifier achieving 97% accuracy, implemented rule-based sentiment analyser with custom financial dictionaries, and integrated these NLP-derived features into deep learning models for market movement forecasting. Developed end-to-end pipeline from raw text to actionable trading signals.

TEACHING & TUTORIALS

HAI101: Human x Artificial Intelligence — Lecture series @ LSE CPNSS (2026), organiser and speaker [**Website**] [**Lecture Recordings**]

Introduction to Transformers and LLMs — Lecture note (2026), with Feng, S. [**PDF**]

Scraping Reddit the Right Way: A Guide to Legal and Ethical Data Collection with RedditHarbor — Tutorial @ ICWSM'24, co-organised with the Open Data Institute (ODI)

EXHIBITIONS

Oh, N., & Park, A. (2026). artefact(s): LeNet-1. *Art Gallery @ the IEEE/CVF Conference on Computer Vision and Pattern Recognition, Denver (**CVPR'26; IEEE TCPAMI Art Award; Spotlight**)* [**Gallery**] [**Video**]

RELEVANT TRAINING & SKILLS

LSE MA314: Algorithms and Programming (First Class)

LSE GV324: Applied Quantitative Methods for Political Science (First Class)

LSE GV249: Research Design in Political Science (Upper Second Class)

LSE Methodology & Quantitative Training: Causal inference and natural experiments in the social sciences (regression, matching, panel data, diff-in-diff, instrumental variables, regression discontinuity)

Python: PyTorch, spaCy, NLTK, transformers, trl, unsloth, vllm, peft, PRAW, OpenAPI